AQA GCSE Chemistry (Separate Science) Unit 2 Bonding, Structure and the Properties of Matter Foundation Revision Activity Mat

a

The three types of chemical bonding are…

1. I

2. C

3. M

Describe the movement and arrangement of subatomic

particles in each of the above.

1.

2.

3.

b

Draw a dot and cross diagram for the following ionic bonding:

sodium chloride

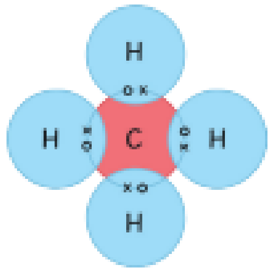
Which four groups are more likely to make ions? Choose from the groups below:

1, 2, 3, 6, 7, 8, 0

d

Using this example, draw dot and cross diagrams for

H_2O, NH_3 and O_2



1. H_2O

c

Describe the bonding in ionic compounds

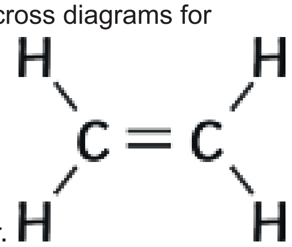
Keywords: ions, negative, positive, opposite,

attraction.

f

Complete the polymer diagram for the following

monomer.



solid

liquid

gas

solution

(g)

(l)

(s)

(aq)

2. NH_3

3. O_2

e

Describe how metals conduct heat and electricity. Use

the diagram to help explain.

h

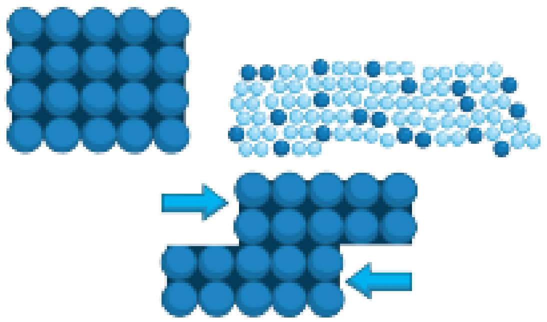
Match up the following with the state symbol.

What is a monomer?

What is a polymer?

g

Properties of metals and alloys.



Describe how the 2 pictures are different to each other.

What happens to the intermolecular forces when a liquid turns into a gas? (Delete the incorrect answers)

Increase

Decrease

Stay the same

Describe the changes of state during:

evaporation:

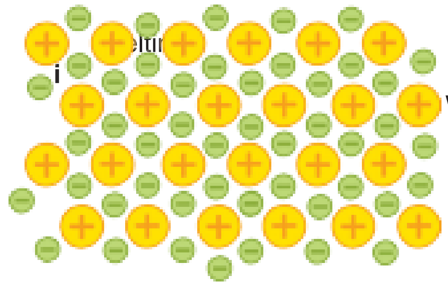
condensation:

j

Small molecules form substances with **high/low** boiling points because they have **strong/weak** intermolecular forces.

They **do/do not** conduct electricity because they do not have any free electrons.

i



Why are some alloys harder than pure metals?

Why can ionic compounds conduct electricity when in solution?

AQA GCSE Chemistry (Separate Science) Unit 2 Bonding, Structure and the Properties of Matter Foundation Revision Activity Mat

a

f

Draw a diagram of the structure of diamond.

Why is this structure so strong? Choose the correct answer.

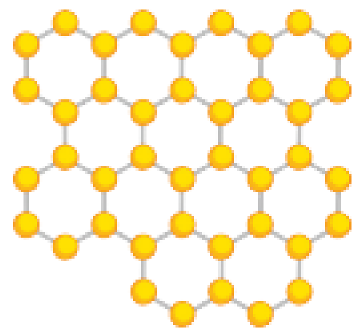
1. Many strong ionic bonds.
2. Many strong covalent bonds.
3. Many strong metallic bonds.

b

i

What are the formulas for the following?

Match up the answers.



_is a single layer of graphite. Why is this material so strong?

j

Iron (II) oxide

Iron (II) hydroxide Iron (III) oxide

$Fe(OH)_2$

FeO

Fe_2O_3

How many:

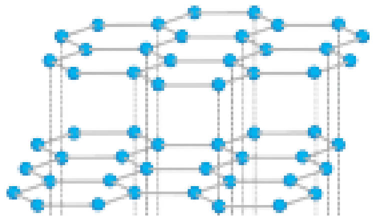
mm in 1m?

m in 1mm?

c

-6 -5
Describe the structure, hardness and conductivity. **Keywords:** covalent, atoms, electricity, electrons, flat

What is this a diagram of?



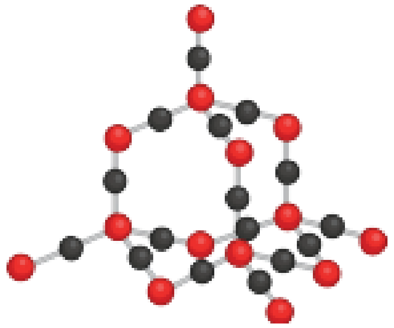
Explain why it can conduct electricity and heat.

Where is this product used?

The study of coarse particles: 2.5 × 10 m to 1 × 10 m diameter. -7 -6

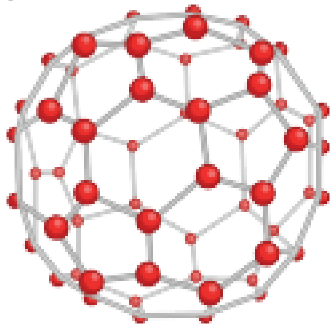
The study of fine particles: 1 × 10 m to 2.5 × 10 m diameter. The study of very small particles: 1 to 100nm diameter.

h
Explain the differences and similarities between silicon dioxide and diamond.



d
Why do the properties of a material made from nanoparticles change when it is in bulk?

g



Name **two** uses of nanoparticles. l Explain why nanoparticles can be potentially harmful to human health. m

What are the abbreviated units for the following:

metre;

centimetre;

millimetre;

nanometre;

micrometre.

e
This is a carbon n_.

It has high t

strength, high h

and e_conductivity.

What is this structure?

How many carbon atoms are there?

a) 20 b) 30 c) 40 d) 50 e) 60

k
Compare diamond and graphite.

What is **nanoscience**? (Tick one answer.)

AQA GCSE Chemistry (Separate Science) Unit 2 Bonding, Structure and the Properties of Matter Foundation Revision Activity Mat Answers 1

a
The three types of chemical bonding are…

1. ionic

2. covalent

3. metallic

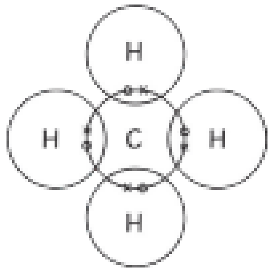
Describe the movement and arrangement

d
Using this example, draw dot and cross diagrams for of subatomic

f
Complete the polymer diagram for the following

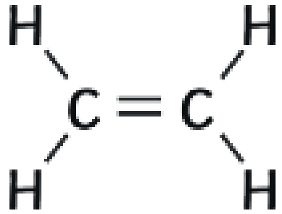
Match up the following with the state symbol.

h



H_2O, NH_3 and O_2

monomer.



solid

liquid

gas

solution

(g)

particles in each of the above.

1. Electrons are lost and gained to fill the outer shell.

2. Electrons are shared to fill the outer shell. 3. Positive metal ions are surrounded by free electrons.

2. NH_3

3. O_2

b

Draw a dot and cross diagram for the following ionic bonding:

sodium chloride

Which four groups are more likely to make ions?

1. H_2O

e

1, 2, 6 and 7

(l)

(s)

(aq)

Describe how metals conduct heat and electricity. Use the diagram to help explain.

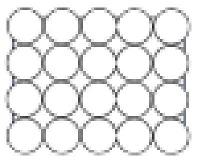
What is a monomer? **One molecule.**

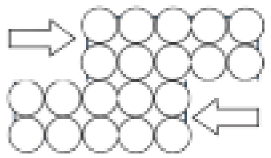
What is a polymer?

A long chain of monomers.

g

Properties of metals and alloys.





Describe the bonding in ionic compounds

c

What happens to the intermolecular forces when a liquid turns into a gas?

Decrease

i

Describe the changes of state during:

evaporation:

liquid changes to a gas.

condensation:

gas changes to a liquid.

melting:

solid changes to a liquid.

j

Small molecules form substances with **low** boiling points because they have **weak** intermolecular forces.

Metals have free

electrons that are able to move around and **transfer energy.**

have any free electrons.

They **do not** conduct electricity because they do not

They are held together by the strong ionic forces of oppositely charged ions. Metal ions have a positive charge and non-metals ions have a negative charge so they are attracted. They have very strong bonds.

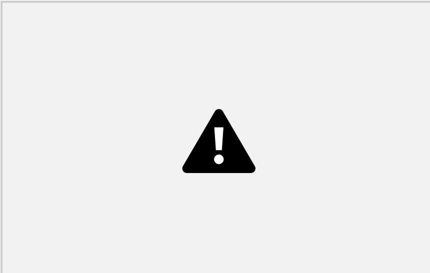
They have different sized particles so the layers can not slide across each other as easily.

Why can ionic compounds conduct electricity when in solution?

The ions are free to move about and can conduct electricity.

Describe how the 2 pictures are different to each other.

Alloys have different sized particles. In pure metals, all the atoms are the same.



Why are some alloys harder than pure metals?

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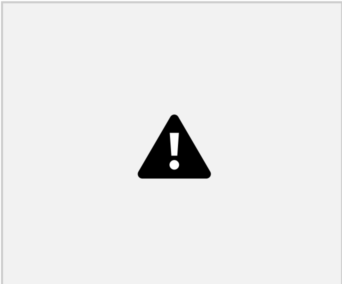
2

AQA GCSE Chemistry (Separate Science) Unit 2 Bonding, Structure and the Properties of Matter Foundation Revision Activity Mat Answers

a

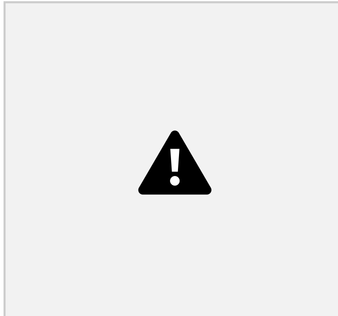
f

Draw a diagram of the structure of diamond.



Why is this structure so strong? Choose the correct answer.

2. Many strong covalent bonds.



Graphene is a single layer of graphite. Why is this material so strong? **It has strong covalent bonds.**

i

What are the formulas for the following?

Match up the answers.

Iron (II) oxide

Iron (II) hydroxide Iron (III) oxide

$Fe(OH)_2$

FeO

Fe_2O_3

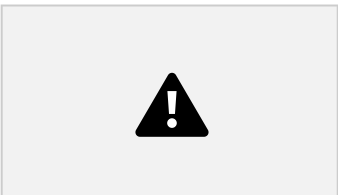
How many:

mm in 1m? **1000mm** m in 1mm? **0.001m**

j

b

What is this a diagram of?



Graphite

Explain why it can conduct electricity and heat.

Graphite has free delocalised electrons that can pass between layers; the electrons can carry the charge.

c

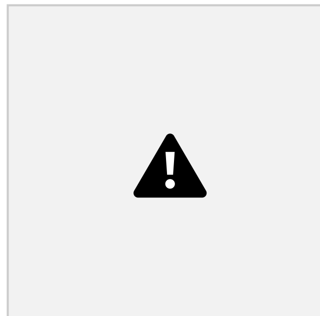
What is **nanoscience**? (Tick one answer.)

The study of coarse particles: 2.5 × 10 m to 1 × 10 m diameter. -6 -5 -7 -6

Where is this product used?

In electronics and composites.

g



What is this structure?

What are the abbreviated units for the following:

metre; **m**

centimetre; c**m**

millimetre; **mm**

nanometre; **nm**

micrometre. **μm**

Compare diamond and graphite.
Describe the structure, hardness and conductivity.

Both – forms of carbon.

Single covalent bonds

Have many atoms.

The study of fine particles: 1 × 10 m to 2.5 × 10 m diameter. **The study of very small particles: 1 to 100nm diameter.**

d

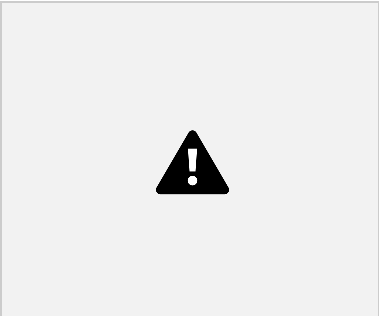
structure to diamond.

e

This is a carbon **nanotube**.

m

Explain why nanoparticles can be potentially harmful to human health.



heat and

The properties of nanoparticles are different to the properties of the same material in bulk. It is difficult for scientists to know how they will behave through studying the material that the nanoparticle has come from.

Why do the properties of a material made from nanoparticles change when it is in bulk?

k This is because of the high surface area to volume ratio.

Buckminsterfullerene

How many carbon atoms are there?

e) 60

h

Explain the differences and similarities between silicon dioxide and diamond. **Silicon dioxide contains silicon and oxygen atoms instead of carbon but has a similar**

Graphite – flat sheets, conducts electricity, each carbon atom forms 3 covalent bonds.

Diamond – tetrahedral structure, each carbon atom forms 4 covalent bonds, does not conduct electricity.

l

Name **two** uses of nanoparticles.

Two from: medicine, electronics, cosmetics, sun creams, deodorants, catalysts.